

COMPOSITES AND ANALYSIS LAB

VII Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AE50	PCC	L	T	P	C	CIA	SEE	Total
		-	-	3	1.5	40	60	100

LIST OF EXPERIMENTS

1. Preparation of fiber reinforced polymer composites using hand- lay-up technique.
 2. Evaluation of flexural properties of polymer based composites
 3. Determine the hardness of fiber reinforced polymers.
 4. Impact test study of polymer based composite materials
 5. Introduction to writing MATLAB code for analysing composite laminate
 6. Analysis of composite sandwich structure – Bending/Buckling
 7. Analysis of Composite Cylindrical Tube
 8. Nonlinear buckling analysis of stiffened plate
 9. Optimization of stiffened plates against buckling
 10. Nonlinear buckling analysis of sub-stiffened plate
 11. Analysis of crack growth in plate
 12. Analysis of composite joints – T joint/I joint
- Note:** Ten experiments should be performed.

Software's Required

ANSYS

Reference Books

1. Barbero, Ever J, Finite Element Analysis of Composite Materials using ANSYS, CRC Press, 2013.
2. Divya Zindani, Apurba Kumafr Roy and Kaushik Kumar, Working with ANSYS: A Tutorial Approach, I K International Publishing House Pvt. Ltd, 2017.

COURSE OUTCOMES : STUDENTS SHOULD BE ABLE TO:

1. Fabricate polymer matrix composite material according to the requirement of mechanical properties
2. Demonstrate testing methods to evaluate mechanical properties of the material
3. Understand the basic features of ANSYS ACP software tools
4. Design and analysis of Aircraft structures fabricated by composite/conventional material
5. Analyse the impact of crack growth/discontinuities in the composite plate